

Pulmonary fibrosis and/or bone marrow failure, telomere-related, 4
Dyskeratosis congenita, autosomal recessive 6

[illegible]

	180	190	200	210	220	230	240	250	260	270																																			
HS	-	CPVTIPEDQKKFIDQV	VEKIEDL	LQSEENKN	LDLEP	-	CTGFQRKLIYQTL	SWKYPKGIHVET	LETEKKERY	IVISKVDEEERKRREQQK																																			
MM	-	CPVTIPEDQKKFIDQV	IEKIEDFL	LQSEEEKRSL	ELDLP	-	CTGFQRKLIYQTL	SWKYPKGIHVET	LETDKKERH	IVISKVDEEERKRREQEK																																			
RN	-	CPVTIPEDQKKFIDQV	IEKIEGFL	QREEKRTL	LELNP	-	CTGFQRKLIYQTL	SWKYPKGIHVET	LETDKKERH	IVISKVDEEERKRREQEK																																			
XT	-	TPVSIPDEQKGFIDKV	VERVEDFL	KNE	-	QKSMDVEP	-	CTGYQRKLIYQTL	NWKYPRG	IEHVETLESEKKERY	IVISKVDEEERKRMEQQK																																		
DR	-	GVCNVPEDQREFIRSV	EЕКVEAL	LKNT	-	DQTL	LDLEP	-	CTGFQRKLIYQTL	NSKYSKGLHVEA	LETEKKERFIQISKVDD	DEERRRREQQK																																	
CE	K	CYQSTSYHITHRT	D	EPMQKIPLKMK	PNSSVSL	LRMPRN	L	SSVEKYM	I	IYEL	T	KAF	P	QFL	-	-	-	-	F	T	C	D	E	K	Q	Q	N	L	H	V	K	N	I	S	D	D	Y	L	K	E	K	D	N	L	E

HS	H A K E Q E E L N D A V G F S R V I H A I A N S G K L V I G H N M L L D V M H T V H Q F Y C P L P A D L S E F K E M T T C V F P R L L D T K L M A S T Q P F K D I I N N T S L A E L
MM	Y T K E Q E E L N D A V G F S R V I H A I A N S G K L V V G H N M L L D V M H T I H Q F Y C P L P A D L N E F K E M A I C V F P R L L D T K L M A S T Q P F K D I I N N T S L A E L
RN	Y A K E Q E E L N D A V G F S R V I H A I A N S G K L V V G H N M L L D V M H T I H Q F Y C P L P V D L N E F K E M T T C V F P R L L D T K L M A S T Q P F K D I I N N T S L A E L
XT	Q A K E R E E L D D A V G F S R I I I Q A I S S S G K L V V G H N M L L D I M H T I H Q F F C Q L P D E L N E F K E V T N C V F P R V L D T K L M A S T N P F K E I I Y N T S L A E L
DR	Q Q R E Q E E L N D A V G F S R V I R A I S K S G K L V V G H N M L L D V M H T I H Q F C G P L P E E L D D F K E V A M T V F P R L L D T K L M A S T Q P F K E I I H N T S L A E L
CE	R A - R A R C S E S V K G V S A I L Q V V H M T G K L V V G H N S L L D A M Y M Y H Y F F S H L P A N Y Q M F K D K F N A L F P R I M D T K L L A Q A L R F E L P G V G D S L E N L
	280 290 300 310 320 330 340 350 360

	370	380	390	400	410	420	430	440	450
HS	EKRLKETP	FNPPKVE	SAEGFPSYDTASEQLHEAGYDAYITGLCFISMANYLGSFLSPPKIHVSAR	SKLIEPFFNKLFLMR					
MM	EKRLKETP	FDPPKVE	SAEGFPSYDTASEQLHEAGYDAYITGLCFISMANYLGSLLSPPKMCVSAR	SKLIEPFFNKLFLMR					
RN	EKRLKETP	FDPPKVE	SAEGFPSYDTASEQLHEAGYDAYITGLCFISMANYLGSLLSPPKTCVSAR	SKLIEPFFNKLFLMR					
XT	EKRLKEAP	FKPPKVD	SAEGFPSYNTASEQLHEAGYDAYITGLCFISMANYLGSFLSPPKGYVSSR	SKIIRPFFNKLFLMR					
DR	HKQLRQKP	FRPPTTE	CPEGLQSYDTSTEQLHEAGYDAFITGLCFISMANYLGSFLTTPPKSHISAR	SKLLEPFYNKLFLMR					
CE	GDYFGSDKSDKTVPPEL	RGFIEPWMNP	LEDESENVYHNAGFDSYVTGEVFLKL	AHIYINRRNNFKNEILDFDRIYQYLEAPILNRL-PFQ					

HS VMDIPYLNLEGPDLQPKRDHVLHVTFPKKEWK--TSDLYQLFSAFG-----NIQISWIDDTSAF-VSLSQPEQVKIAVNNTSKYAESYRIQ
 MM VMDIPYLNLEGPDLQPKRDHVLHVTFPKKEWK--TSDLYQLFSAFG-----NIQISWIDDTSAF-VSLSQPEQVQIAVNNTSKYAESYRIQ
 RN VMDIPYLNLEGPDLQPKRDHVLHVTFPKKEWK--TSDLYQLFSAFG-----NIQISWIDDTSAF-VSLSQPEQVQIAVNNTSKYAESYRIQ
 XT IMDIPYLNLEGPDLQPKRDNVLHVAFPKKEWK--TSDLYQLFSAFG-----NIQVSWIDDTSAF-VSLSHPDQVQIAVNNTSKYAESYRIQ
 DR VIDIPYLNMSGPDLPKRDHVLVYVTFPKKEWK--TSDLYQLFSAFG-----NIQVSWVDDTSAF-VSLSQTEQVQIAMNTSRYAESYRIQ
 CE LMDVGCCYLTGDDSKGFRPDVITIVRRDRVAIEEDEFRLYLEKALGTLMATYQFDIEWSKNKKELFLATNSPGSYA--FLCEKFFSNDSSLA

HS TYAEYMGRKQEEKQIKRKWTED - - - - SWKEADSKRLNPQCIPYTLQNHYYRNN SFTAPSTVGKRNLSPSQEEAGLEDGVS - - - - -
 MM TYAEYVGKKQEGKQVKRKWTED - - - - SWKEVDKRKRP H - - - - - MQGPCYHSNSFTAAGVLGKRTLSPDPREAALEDRES - - - - -
 RN TYAEYVGKKQESKQVKRKWTED - - - - SWKEVDKRKRP H - - - - - VPGPCYHTNSFTAAGIVGKRTLSPDPREAGVVDRES - - - - -
 XT TYAEYVEKKNEENQTKRKWAED - - - - GWKDLERKRLKTQYNSYIPQTPVFYGNCFVAPS YAVKRSMSP IQEETASENTEE - - - - -
 DR TYAEYLQSRQKNTHSSRKWASD - - - - GWADTSYPSVAMTTASG - - - - - YSHTDNW - - - - - HQAVKRSISP SLDEQNHGADS SWTNYSVKKIK
 CE - - - - - - - - PLDELD SGK KWT FEQRQTAWRSFKNRGV GIGINAKRIRA - - - - - QNQ - - - - - ATKIQ RAMEI - - - - -

HS --GEISDTELEQTDSCAE-----PLSEGRKKAKKKLKRMMKKEKLS PAGSISKNSPATLFEVPDTW
 MM --EEVSDSELEQTDSCD-----PLPEGRKKSKKKLKRMMKKEKLSLAGSVS--DSPAVLFEVPDTW
 RN --EGISDPELEQTDYCAD-----PLPEGRKKAKKKLKRMMKKEFSLAGSAS--DSPAVLFEVPDTW
 XT --VRAQE-----DDPSNP-----GATEQGKKPKKNHKRQKTDSTPPETSDSGSSVLFEPVDTW
 DR TEGSCTQTYADVAGSCDWPRLLQADEGGASVSPVAEEAELDEFSSANQSQGKRSRKKKK-----RKSDASETTPPALFDVPPQVW
 CE -----